Kingdom of Saudi Arabia
Ministry of Education
Umm Al-Qura University
031

المملكة العربية السعودية
وزارة التعليم
جامعة أم القرى
٠٣١

الاختبار النصفي لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

مدة الاختبار	تاريخ الاختبار	الفصل الدراسي
ساعة	الأربعاء 1447/10/20 الموافق 2026/04/08	472

تعليمات الاختبار:

- عدد أسئلة الاختبار ١٢ سؤالا.
- درجة الاختبار ٣٠ درجة،
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة .
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة
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Exam Instructions

- The exam consists of **12 questions**.
- The total exam score is **30 marks**.
- Make sure to clearly write your **name, university ID number, and section number** on the answer sheet.
- **No notes or reference materials are allowed** during the exam (**closed-book exam**).
- At the end of the exam, **submit both the question paper and the answer sheet** to the invigilator.
- If you need assistance or have a question, **raise your hand and wait for the invigilator** to assist you.
- Please **manage your time carefully** to ensure you are able to answer all required questions.

رقم الشعبة	اسم الطالب	الرقم الجامعي
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1. What are the three main parts that make up the Transformer architecture?

1- ~~Tokenizer~~ ✗
2- Transformer Blocks Stack
3- Language Modeling Head

[illegible]

Encoder
Decoder
Attention

2. What is the primary function of "Positional Encoding" in Transformers?

Since attension doesn't care about words' order and positions, transformers use Positional Encoding to track each word's(token) position. ✓

ممتاز هنا وفهمك تام لل PE

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

1- Query
2- Key
3- Value ✓

4. In the self-attention mechanism, what does the Query (Q) represent?

A) The stored information retrieved as a result of the attention process

B) The current word that is "asking" which other words are most relevant to it ✓

C) A label used to index and match words in the attention lookup table

D) The scaling factor applied before the SoftMax to stabilize gradients

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

A) True ✓

B) False

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6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words

B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words

C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization

D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

A) To normalize the values and prevent large numbers

B) To make the matrix dimensions compatible for multiplication

C) To reduce the size of the matrix for faster computation

D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

"Attention is all you need" paper, designed to perform translation task.

9. In Encoder-Decoder Attention, where do Q, K, V come from?

While K and V come from encoder, Q comes from decoder.

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only Transformers?

1-RAG

2-Text Classification

هذه الإجابة تدل على حديث للتطبيقات
وانك فاهم يا مصعب مش حافظ
الله يوفقك

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11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

Word Embedding is the process of translating words (tokens) to vectors. Positional Encoding is the process of tracking each word's position.
They're complementary processes, while the vectors of the Word Embedding captures the meaning of a single token, Positional Encoding add position and order information to capture the meaning of the full sentence.

12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	Self-attention	Masked self-attention
Goal	Represent and understand contextual relations	Predict the next token based on the previous ones
Example tasks	Text Classification	Question-Answering

Why ChatGPT is generative:

It's decoder only model, so its task is to predict (generate) the next token.

تبارك الرحمن
مصعب إنت ممتاز
الله يوفقك ويزيدك من فضله

~ Good Luck ~

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Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		**30**